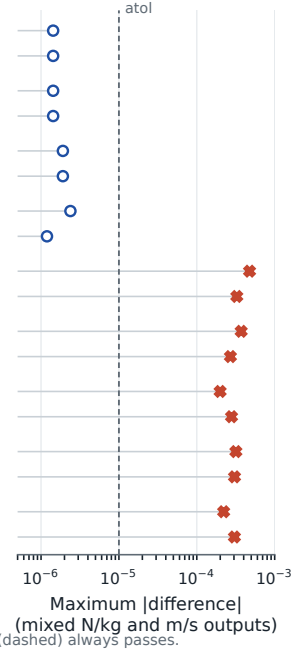
a Route-level qualification: nine families × two repeats

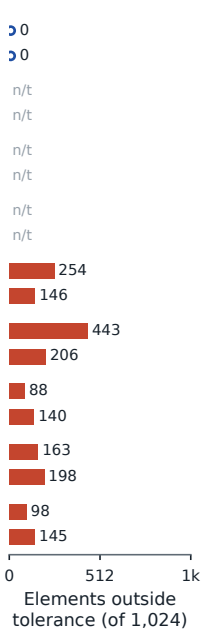
Recovered CPU vs original CPU: 18/18 exactly equal · Recovered CPU vs archived GPU arrays: 8/18 within tolerance, 10/18 outside

Route	Recovered CPU = original CPU	Archived GPU comparison
M0 · r0	■ exact	○ within tolerance
M0 · r1	■ exact	○ within tolerance
M1 · r0	■ exact	○ within tolerance
M1 · r1	■ exact	○ within tolerance
M2 · r0	■ exact	○ within tolerance
M2 · r1	■ exact	○ within tolerance
M3 · r0	■ exact	○ within tolerance
M3 · r1	■ exact	○ within tolerance
M4 · r0	■ exact	✖ outside tolerance
M4 · r1	■ exact	✖ outside tolerance
M5 · r0	■ exact	✖ outside tolerance
M5 · r1	■ exact	✖ outside tolerance
M6 · r0	■ exact	✖ outside tolerance
M6 · r1	■ exact	✖ outside tolerance
M7 · r0	■ exact	✖ outside tolerance
M7 · r1	■ exact	✖ outside tolerance
M8 · r0	■ exact	✖ outside tolerance
M8 · r1	■ exact	✖ outside tolerance

b Largest difference



c Failing elements



Criterion per element: $|a - b| \le 10^{-5} + 10^{-4}|b|$ (a replayed, b archived). A difference ≤ atol = 10^{-5} (dashed) always passes.
n/t: failure count not tabulated. Recorded private comparison summarized publicly; no GPU replay or new inference was run for this article.